

README File

a. Setup Steps

i. OS / Compiler Version

- Tested on Linux (Ubuntu 22.04 / Arch Linux).
- Requires g++ (≥ 9.0) with C++11 or later.

ii. Dependencies

- Standard C++ libraries only.
- No external dependencies.
- Compile using g++ directly.

b. How to Run

Part 1(A): Without Cache

```
Compile:
g++ 1aS.cpp -o server1A -pthread
g++ 1aC.cpp -o client1A -pthread

Run Server:
./server1A    (listens on port 19036)

Run Client(s):
./client1A    (connects to server at port 19036)
```

Part 1(B): With Cache

```
Compile:
g++ 1bS.cpp -o server1B -pthread
g++ 1bC.cpp -o client1B -pthread

Run Server:
./server1B    (listens on port 13036)

Run Client(s):
./client1B    (connects to server at port 13036)
```

ii. Default Parameters

- Port:
 - Part 1(A): 19036
 - Part 1(B): 13036
- Timeout: Keep-alive pings every ~5 seconds
- Number of Questions: 10 per genre

c. Expected Input/Output Format

Sample Run (Server + Client)

```
Server (1A) Output:
[INFO] Parsed 10 questions.
[KEEP_ALIVE] Sent ping -> socket 5
[SERVER] Generating quiz for genre: science
```

```
Client (1A) Session:
■ Connected to Quiz Server (port 19036)
Commands:
genre <name>
ans <A-D>
```

```
    lead
    exit

> genre science
--- QUESTION ---
Which planet is closest to the Sun?
A) Earth
B) Mercury
C) Venus
D) Mars
-----
ans B
> [CLIENT] Correct
```

Leaderboard Output

```
Client Request:
> lead

Server Response:
--- LEADERBOARD ---
Client4:1
Client5:0
Client6:0
-----
```

Part 1(B) Cache Behavior Example

```
Client Input:
> genre history

Server Response:
INFO|Cached questions used for genre history
--- QUESTION ---
Who was the first US president?
A) Lincoln
B) Washington
C) Jefferson
D) Adams
-----
ans B
> [CLIENT] Correct
```